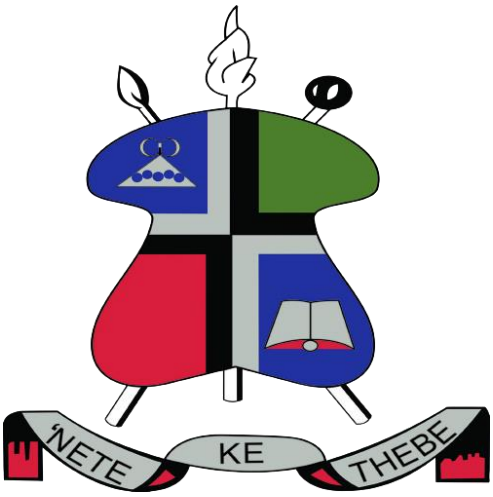


National University of Lesotho
Faculty of Science and Technology
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Requirements Engineering - Phase 2

Learning Management System

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1. Stakeholder analysis and conflict resolution:

1.1 Identified stakeholders and their interests

A stakeholder is a person or group that affects, uses, manages, or is affected by the system. The below table is a list of stakeholders and their main interests to the system.

Stakeholders	**Main Interests**
Student	View enrolled courses and assignments, submit and resubmit work before deadlines, view released marks and written feedback.
Lecture	Create assignments, set deadlines and maximum marks, grant individual extensions, mark submissions, provide feedback, and see marks promptly after marking.
System Administrator	Manage user accounts, departments and courses, assign lecturers, enroll students, and protect credentials and records.
Principal	Ensure the system is simple for teachers who are not IT experts, prevent students from viewing marks before HoD approval.
Head of Department	Review and moderate marks before release, ensure assessment rules are applied consistently and maintain assessment integrity.
IT Officer	Ensure compatibility with older computers, offline operation, acceptable performance for records involving 800 students, and password protection.
Developer	Deliver a working, tested C++ application within the seven-week schedule, using the specified tools and following the Waterfall model with V-Model verification and validation

1.2 Conflicts and ambiguities in stakeholder statements

The below table identifies the conflicting statements, provide a resolution, and state the resulting requirement or business rule.

Conflict / ambiguity	Resolution	Resulting business rule
Lecturer visibility vs student visibility: Lecturers want to see marks as soon as they are marked, while the principal insists that students must not see marks before HoD approval.	Separate the marking stage from the release stage. Lecturers can view marks they have entered and students can view marks only after HoD moderation and release.	A mark must not be visible to a student until it has been approved by the HoD and released.
Late submissions vs automatic rejection: The student representative argues that late work should not simply be rejected, while the HoD specifies a late penalty.	Accept late submissions within the permitted late-submission window and apply the specified penalty rather than rejecting them automatically.	A submission received after the deadline is late. Deduct 10% of the assignment's maximum available mark per day, up to three days. After three days, the mark is zero.
Resubmission vs deadline enforcement: Students want to correct mistakes, but deadlines must still be enforced.	Permit resubmission before the applicable deadline. Each valid resubmission becomes the current submission for marking and the system should retain an audit history where required.	Students may resubmit before the applicable deadline. The latest valid submission received before that deadline is the submission to be marked.
Individual extensions vs a common deadline: Lecturers need to grant extensions to individual students, so a single assignment deadline cannot apply to every student in all circumstances.	Allow authorized lecturers to grant and record an individual extension. The applicable deadline is the original deadline unless a valid extension exists.	An approved extension changes the deadline for the specified student only and must record the revised deadline and authorizing lecturer.
Lecturer marking vs HoD moderation: A lecturer may regard marking as complete, but the HoD must review every mark before release.	Use separate submission states for marking, moderation and release. Marking does not automatically make a mark visible to students.	Only marks that have passed HoD moderation may be released to students.
Simplicity vs security: Teachers need a simple system, but student information and assessment records must be protected	Provide role-specific menus and straightforward workflows while enforcing permissions in the application logic.	Each user may perform only operations authorized for their role. Passwords must be stored as salted hashes, not plaintext.
Internet reliability vs system availability: The IT officer says the system must work offline, despite unreliable Internet access	Design the application and database to operate locally without depending on an Internet connection.	Core account, course, assignment, submission, marking and moderation functions must work on the local installation without Internet access.

2. Software Requirements Specification (SRS):

2.1 Purpose and scope

System name: Moderated Learning Management System (MLMS)

Purpose: This Software Requirements Specification defines the functional and non-functional requirements of the Moderated Learning Management System. It serves as the agreed baseline for system design, implementation, verification, and validation.

Scope: The system will support students, lecturers, HoDs and system administrators. It will be implemented in C++17, use SQLite for data storage, and operate on a single school computer or a local network. Core operations must not depend on Internet access.

Out of scope: Online payments, video conferencing, automated grading, plagiarism detection, and web/mobile interfaces.

2.2 Functional requirements

Functional requirements describe the specific behaviors, services, functions, and tasks that a software system must provide to satisfy a user's needs.

The priorities: “Must” means essential and have to be implemented, “Should” means important but not essential for the operation of the system and “Could” means may be implemented if time permits.

ID	Functional requirement	Priority	Verification method
FR1	The system shall authenticate users using a registered username and password before granting access to protected functions.	Must	Authentication tests using valid and invalid credentials.
FR2	The system shall restrict each authenticated user to functions permitted by their assigned role.	Must	Role-based access-control tests for all four roles.
FR3	The system shall allow the administrator to create, update, deactivate, and manage user accounts.	Must	Account management functional tests.
FR4	The system shall allow the administrator to create and manage departments and courses.	Must	Create, update, and retrieval tests.
FR5	The system shall allow the administrator to assign lecturers to courses and enroll students in courses.	Must	Assignment and enrolment tests.
FR6	The system shall allow lecturers to create assignments with a title, description, deadline, and maximum mark.	Must	Create-assignment tests with valid and invalid inputs.

FR7	The system shall allow students to view their enrolled courses and the assignments associated with those courses.	Must	Student-access and course-visibility tests.
FR8	The system shall allow students to submit assignment work and resubmit it before the applicable deadline.	Must	Submission and resubmission tests.
FR9	The system shall record the submission time and determine whether each submission is on time or late.	Must	Boundary tests immediately before, at, and after the deadline.
FR10	The system shall calculate the late penalty as 10% of the available mark per late day for up to three days, and assign zero for work submitted more than three days late, subject to an approved extension.	Must	Tests for each late-day boundary and extension scenario.
FR11	The system shall allow lecturers to grant an individual student an extension and record the revised deadline.	Must	Extension tests verifying that only the affected student's deadline changes.
FR12	The system shall allow lecturers to record marks and written feedback for submitted assignments.	Must	Marking tests using valid marks and feedback.
FR13	The system shall allow an HoD to review and moderate marks for courses within their department, including approving or returning marks for correction.	Must	Moderation workflow and department-authorization tests.
FR14	The system shall prevent students from viewing marks until the HoD has approved and released them.	Must	Access tests before and after release.
FR15	The system shall allow students to view released marks and associated written feedback.	Must	Tests for released and unreleased results.
FR16	The system shall record each mark change with the responsible user, timestamp, previous mark, new mark, and reason for the change.	Must	Audit-trail tests that inspect stored records after mark changes.

2.3 Non-functional requirements

Non-functional requirements define measurable quality attributes or operating constraints.

ID	Non-functional requirement	Priority	Verification method
NFR1 Performance	With records for 800 students stored in the database, 95% of assignment-list and released-result retrieval requests shall complete within 2 seconds under the agreed test conditions.	Must	Performance test using a representative 800-student dataset.
NFR2 Authentication	The system shall store passwords only as salted password hashes; plaintext passwords shall not be stored in the database or audit logs.	Must	Database inspection and authentication/security tests.
NFR3 Access control	In the defined role authorization test suite, 100% of unauthorized attempts to access restricted functions shall be denied.	Must	Automated permission tests for all four roles.
NFR4 Data integrity	After 100 consecutive successful create, update, and retrieval operations in the persistence test suite, all saved records shall match the expected values, with no unexplained loss or corruption.	Must	Automated SQLite persistence tests.
NFR5 Offline operation	The core workflows login, course and assignment retrieval, submission recording, marking, moderation, and result release shall complete without an internet connection on the supported school computer or local network.	Must	Execute the defined workflows with internet access disconnected.
NFR6 Usability	In a usability test, at least 4 out of 5 representative lecturer users shall complete assignment creation, marking, and feedback entry using the user guide, without assistance, within 5 minutes per task.	Should	Observed usability test with representative users.
NFR7 Auditability	For 100% of successful mark change operations in the audit test suite, the system shall retain the acting user, timestamp, previous mark, new mark, and reason.	Must	Automated audit completeness tests.
NFR8 Maintainability	The business-logic module shall achieve at least 70% line coverage in the automated unit-test suite.	Should	Generate and inspect a code coverage report.

2.4 Business Rules

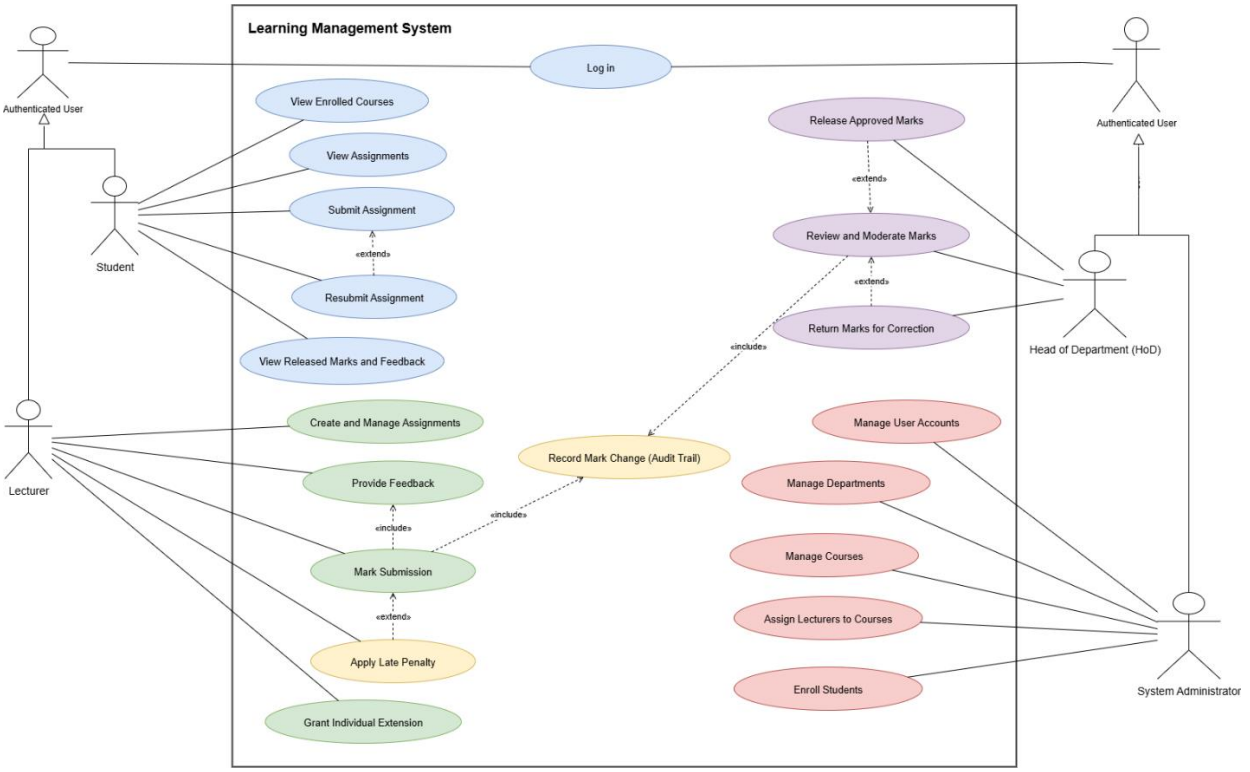
1. Students must not see marks before the HoD has moderated and released them.
2. Late submissions incur a penalty of 10% of the maximum available mark per day, for up to three days; submissions later than three days receive zero, subject to an approved extension.
3. Students may resubmit before their applicable deadline.
4. An individual extension applies only to the student for whom it was approved.
5. The HoD must review marks before they can be released.
6. Every mark change must record who made the change, when it occurred and why.
7. Users may perform only operations permitted by their assigned roles.

3. Use case model:

The below is a table of actors and their main use cases.

Actor	**Main use cases**
Student	Log in, view enrolled courses, view assignments, submit assignment, resubmit assignment, view released marks and feedback.
Lecturer	Log in, manage assignments, grant individual extension, mark submission, provide feedback.
HoD	Log in, review marks, moderate marks, approve marks, return marks for correction, release approved marks.
System Administrator	Log in, manage user accounts, manage departments and courses, assign lecturers, enroll students.

3.1 Use Case Diagram



3.2 Use Case: Submit Assignment

Primary actor: Student

Goal: Allow a student to submit assignment work, or resubmit revised work when permitted, and receive confirmation that the submission has been recorded.

Field	**Specification**
Primary actor	Student
Supporting actor	None required
Trigger	The student selects an assignment and chooses to submit work.
Preconditions	1. The student is authenticated. 2. The student is enrolled in the course associated with the assignment. 3. The assignment exists and is available for submission.
Postconditions on success	1. The submission is saved persistently. 2. The submission timestamp and status are recorded. 3. The student receives confirmation. 4. If this is a resubmission, the system records the revised submission according to the submission-history policy.
Failure postcondition	If submission fails, the system reports the error and does not falsely confirm that the work was saved.

Main success flow

1. The student logs in to the LMS.
2. The student views their enrolled courses.
3. The student selects a course.
4. The system displays the assignments available for that course, including their deadlines.
5. The student selects the assignment to submit.
6. The system displays the assignment details and applicable deadline.
7. The student selects the work to submit and confirms the submission.
8. The system validates the submission and checks the applicable deadline.
9. The system records the submission, timestamp, and submission status.
10. The system displays a confirmation message to the student.

Alternative flows

Student resubmits work

1. At Step 7, the student chooses to submit a revised version of previously submitted work.
2. The system checks whether resubmission is permitted under the applicable deadline and rules.
3. If permitted, the system records the new submission and its timestamp.
4. The system confirms the resubmission.

Submission is late but within the permitted period

1. At Step 8, the system determines that the submission is after the applicable deadline.
2. The system checks whether the submission falls within the three-day late-submission period.
3. The system records the submission as late.
4. The system informs the student that the late-submission policy applies.
5. The submission proceeds to marking.

Student has an approved extension

1. At Step 8, the system checks the student's applicable deadline.

2. The system identifies an approved individual extension.
3. The system evaluates the submission against the revised deadline.
4. The system records the submission status based on the revised deadline.

Exception flows

Student is not enrolled

1. The system determines that the student is not enrolled in the course.
2. The system denies the submission.
3. The system displays an appropriate error message.

Submission is more than three days late

1. The system determines that the submission exceeds the permitted late period, using the applicable deadline.
2. The system informs the student that the late-submission policy assigns zero for work beyond the permitted period.
3. The system applies the agreed policy for recording such submissions and their marks.

Submission cannot be saved

1. The system encounters a storage or validation error.
2. The system informs the student that the submission was unsuccessful.
3. The system does not display a successful-submission confirmation.

Business rules

1. Only authenticated students enrolled in the relevant course may submit work for its assignments.
2. Students may resubmit before the applicable deadline.
3. An approved individual extension changes the applicable deadline for that student.
4. Late work is penalized by 10% of the assignment's maximum mark per late day, for up to three days; work submitted after that period receives zero, subject to the approved-extension rule.
5. Submission timestamps and statuses must be recorded accurately.
6. Students cannot view marks until the HoD has approved and released them.

3.3 Use Case: Mark Submission

Primary actor: Lecturer

Goal: Allow an authorized lecturer to assess a student's submission, record a mark, and provide written feedback for subsequent moderation by the HoD.

Field	Specification
Primary actor	Lecturer
Supporting actor	HoD, during the later moderation stage
Trigger	The lecturer selects a submission that requires marking.
Preconditions	1. The lecturer is authenticated. 2. The lecturer is assigned to the relevant course. 3. The assignment and submission exist. 4. The submission is available for marking.
Postconditions on success	1. The mark is saved. 2. Written feedback is saved. 3. The submission status is updated to marked. 4. The mark is available for HoD moderation but is not yet visible to the student.
Failure postcondition	If the mark cannot be saved, the system reports the failure and does not falsely indicate that marking is complete.

Main success flow

1. The lecturer logs in to the LMS.
2. The lecturer selects a course assigned to them.
3. The system displays the assignments and their submissions.
4. The lecturer selects a submission to mark.
5. The system displays the submitted work, assignment maximum mark, applicable deadline, submission status, and any relevant extension.
6. The lecturer assesses the work.
7. The lecturer enters a mark within the permitted range and provides written feedback.
8. The system validates the mark and required information.
9. The system calculates the applicable late penalty where necessary.
10. The system saves the mark and feedback.
11. The system records the mark change in the audit trail, including the responsible user, timestamp, previous mark where applicable, new mark, and reason.

12. The system updates the submission status to marked.
13. The system informs the lecturer that the mark has been saved and is awaiting moderation.

Alternative flows

Submission is late

1. At Step 5, the system identifies that the submission is late.
2. The system calculates the applicable penalty according to the agreed late-submission rules.
3. The lecturer marks the work in accordance with the policy.
4. The system validates and records the resulting mark and feedback.

Lecturer corrects an existing mark

1. The lecturer opens a previously marked submission that is available for correction.
2. The lecturer enters the corrected mark and the reason for the change.
3. The system validates the revised mark.
4. The system records the old mark, new mark, timestamp, lecturer identity, and reason in the audit trail.
5. The system saves the revised mark for moderation.

Exception flows

Invalid mark

1. At Step 8, the system detects a mark below zero or above the assignment's maximum mark.
2. The system rejects the mark.
3. The system displays an error explaining the permitted range.
4. The lecturer enters a valid mark.

Lecturer lacks permission

1. The system determines that the lecturer is not assigned to the course or is not authorised to mark the submission.
2. The system denies the operation.
3. The system records the event if required by the security logging policy and displays an appropriate message.

Database or save failure

1. The system fails to save the mark or feedback.
2. The system reports that the operation was unsuccessful.
3. The system preserves the integrity of the stored assessment record and allows the lecturer to retry.

Business rules

1. Only an authorized lecturer may mark submissions for their assigned courses.
2. A mark must be between zero and the assignment's maximum mark.
3. Written feedback must be recorded with the mark.
4. The applicable late-submission policy must be enforced consistently.
5. Every mark change must be auditable, including the user, timestamp, previous mark, new mark, and reason.
6. A lecturer's mark is not released to students until the HoD approves and releases it.

3.4 Use Case: Moderate and Release Marks

Primary actor: Head of Department (HoD)

Goal: Ensure that marks are reviewed and approved by the HoD before they become visible to students, with the ability to return marks for correction.

Field	**Specification**
Primary actor	Head of Department (HoD)
Supporting actors	Lecturer and Student, indirectly
Trigger	The HoD selects a course or assignment with marks awaiting moderation.
Preconditions	1. The HoD is authenticated. 2. The HoD is authorised for the relevant department. 3. The lecturer has saved marks for the submissions being reviewed.
Postconditions on approval and release	1. The reviewed marks are approved. 2. The release action is recorded. 3. Students can view the released marks and feedback.
Postconditions on return	The affected marks remain unreleased, and the relevant submissions are returned to the lecturer for correction.
Failure postcondition	If moderation or release fails, the system does not expose unapproved marks to students.

Main success flow

1. The HoD logs in to the LMS.
2. The HoD selects a department and an assignment requiring moderation.
3. The system displays the submissions, lecturer-entered marks, written feedback, and relevant assessment information.
4. The HoD reviews the marks and feedback.
5. The HoD decides that the marks are acceptable.
6. The HoD approves the marks.
7. The HoD selects the option to release the approved results.
8. The system verifies that the marks have been approved and that the HoD is authorised to release them.
9. The system records the release action and updates the relevant submission statuses.
10. The system makes the released marks and feedback available to the relevant students.
11. The system confirms that the results have been released.

Alternative flows

HoD returns marks for correction

1. At Step 4, the HoD identifies a mark or feedback entry requiring correction.
2. The HoD selects the affected submission and records the reason for returning it.
3. The system records the moderation outcome.
4. The system returns the submission to the lecturer for correction.
5. The mark remains unreleased.
6. The lecturer corrects the mark or feedback and resubmits it for moderation.

HoD approves marks but delays release

1. At Step 5, the HoD accepts the marks.
2. The HoD approves the marks but does not choose to release them immediately.
3. The system records the approval status.
4. The marks remain hidden from students until the release action is completed.

Exception flows

HoD lacks authority

1. The system determines that the HoD is not authorised to moderate marks for the selected department.
2. The system denies the operation.
3. The system displays an access-denied message.

Marks are incomplete

1. During Step 3 or Step 4, the system identifies that one or more required marking records are incomplete.
2. The system identifies the affected records.
3. The system prevents completion of the moderation or release operation for records that do not satisfy the required conditions.
4. The HoD or lecturer resolves the issue before the workflow continues.

Release operation fails

1. The system fails to complete the release operation.
2. The system reports the failure to the HoD.
3. The system ensures that students cannot view marks that have not been successfully released.
4. The HoD may retry the release operation.

Business rules

1. Only an authorised HoD may moderate and release results for courses within their department.
2. All marks must undergo moderation before release to students.
3. Marks returned for correction must remain hidden from students until they are approved and released.
4. Approval and release are distinct actions; approval alone does not make results visible.
5. Mark changes must be recorded in the audit trail.
6. Students may view only the marks and feedback that have been released to them.